

- 8 LibertyLink® soybean is a genetically modified crop. It was first grown in 1996 and used in food products from 1998. It has been grown in 6 countries and used in food products in 21 countries.

Table 8.1 summarises the modifications made to the soybean plant to produce LibertyLink® soybean.

Table 8.1

name of introduced gene	gene donor organism	gene product	function of gene product
<i>pat</i>	<i>Streptomyces viridochromogenes</i>	phosphinothricin N-acetyltransferase	stops action of glufosinate, a herbicide

- (a) Explain how the modification made to produce LibertyLink® soybean may help to solve the global demand for food.

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- (b) (i) Name the type of enzyme that could be used to cut out the *pat* gene from *S. viridochromogenes*.

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- (ii) Name the enzyme that could be used to join the *pat* gene to a plasmid by forming phosphodiester bonds.

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- (c) Suggest reasons why LibertyLink® soybean is used in food products in 21 countries but only grown in 6 countries.

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[Total: 9]